

绪论

肖亮海

暨南大学 信息科学技术学院

什么是量化交易？



CLMS **45.46** **+11** 1:45.46/45.47 38x14 Prev 45.33
AT 11:08d Vol 276515 Op 45.30 Hi 45.71 Lo 44.94 OpenInt 501015
CLMS COMB Comdty 0 Actions 0 Settings Futures Contract Table
WTI CRUDE FUTURE Contracts 73 As of 07/28/16 Session 1,726,939
1st/2nd/3rd/4th/5th/6th/7th/8th/9th/10th/11th/12th/13th/14th/15th/16th/17th/18th/19th/20th/21st/22nd/23rd/24th/25th/26th/27th/28th/29th/30th/31st/32nd/33rd/34th/35th/36th/37th/38th/39th/40th/41st/42nd/43rd/44th/45th/46th/47th/48th/49th/50th/51st/52nd/53rd/54th/55th/56th/57th/58th/59th/60th/61st/62nd/63rd/64th/65th/66th/67th/68th/69th/70th/71st/72nd/73rd/74th/75th/76th/77th/78th/79th/80th/81st/82nd/83rd/84th/85th/86th/87th/88th/89th/90th/91st/92nd/93rd/94th/95th/96th/97th/98th/99th/100th/101st/102nd/103rd/104th/105th/106th/107th/108th/109th/110th/111th/112th/113th/114th/115th/116th/117th/118th/119th/120th/121st/122nd/123rd/124th/125th/126th/127th/128th/129th/130th/131st/132nd/133rd/134th/135th/136th/137th/138th/139th/140th/141st/142nd/143rd/144th/145th/146th/147th/148th/149th/150th/151st/152nd/153rd/154th/155th/156th/157th/158th/159th/160th/161st/162nd/163rd/164th/165th/166th/167th/168th/169th/170th/171st/172nd/173rd/174th/175th/176th/177th/178th/179th/180th/181st/182nd/183rd/184th/185th/186th/187th/188th/189th/190th/191st/192nd/193rd/194th/195th/196th/197th/198th/199th/200th/201st/202nd/203rd/204th/205th/206th/207th/208th/209th/210th/211st/212nd/213th/214th/215th/216th/217th/218th/219th/220th/221st/222nd/223rd/224th/225th/226th/227th/228th/229th/230th/231st/232nd/233rd/234th/235th/236th/237th/238th/239th/240th/241st/242nd/243rd/244th/245th/246th/247th/248th/249th/250th/251st/252nd/253rd/254th/255th/256th/257th/258th/259th/260th/261st/262nd/263rd/264th/265th/266th/267th/268th/269th/270th/271st/272nd/273rd/274th/275th/276th/277th/278th/279th/280th/281st/282nd/283rd/284th/285th/286th/287th/288th/289th/290th/291st/292nd/293rd/294th/295th/296th/297th/298th/299th/300th/301st/302nd/303rd/304th/305th/306th/307th/308th/309th/310th/311st/312nd/313th/314th/315th/316th/317th/318th/319th/320th/321st/322nd/323rd/324th/325th/326th/327th/328th/329th/330th/331st/332nd/333rd/334th/335th/336th/337th/338th/339th/340th/341st/342nd/343rd/344th/345th/346th/347th/348th/349th/350th/351st/352nd/353rd/354th/355th/356th/357th/358th/359th/360th/361st/362nd/363rd/364th/365th/366th/367th/368th/369th/370th/371st/372nd/373rd/374th/375th/376th/377th/378th/379th/380th/381st/382nd/383rd/384th/385th/386th/387th/388th/389th/390th/391st/392nd/393rd/394th/395th/396th/397th/398th/399th/400th/401st/402nd/403rd/404th/405th/406th/407th/408th/409th/410th/411st/412nd/413th/414th/415th/416th/417th/418th/419th/420th/421st/422nd/423rd/424th/425th/426th/427th/428th/429th/430th/431st/432nd/433rd/434th/435th/436th/437th/438th/439th/440th/441st/442nd/443rd/444th/445th/446th/447th/448th/449th/450th/451st/452nd/453rd/454th/455th/456th/457th/458th/459th/460th/461st/462nd/463rd/464th/465th/466th/467th/468th/469th/470th/471st/472nd/473rd/474th/475th/476th/477th/478th/479th/480th/481st/482nd/483rd/484th/485th/486th/487th/488th/489th/490th/491st/492nd/493rd/494th/495th/496th/497th/498th/499th/500th/501st/502nd/503rd/504th/505th/506th/507th/508th/509th/510th/511st/512nd/513th/514th/515th/516th/517th/518th/519th/520th/521st/522nd/523rd/524th/525th/526th/527th/528th/529th/530th/531st/532nd/533rd/534th/535th/536th/537th/538th/539th/540th/541st/542nd/543rd/544th/545th/546th/547th/548th/549th/550th/551st/552nd/553rd/554th/555th/556th/557th/558th/559th/560th/561st/562nd/563rd/564th/565th/566th/567th/568th/569th/570th/571st/572nd/573rd/574th/575th/576th/577th/578th/579th/580th/581st/582nd/583rd/584th/585th/586th/587th/588th/589th/590th/591st/592nd/593rd/594th/595th/596th/597th/598th/599th/600th/601st/602nd/603rd/604th/605th/606th/607th/608th/609th/610th/611st/612nd/613th/614th/615th/616th/617th/618th/619th/620th/621st/622nd/623rd/624th/625th/626th/627th/628th/629th/630th/631st/632nd/633rd/634th/635th/636th/637th/638th/639th/640th/641st/642nd/643rd/644th/645th/646th/647th/648th/649th/650th/651st/652nd/653rd/654th/655th/656th/657th/658th/659th/660th/661st/662nd/663rd/664th/665th/666th/667th/668th/669th/670th/671st/672nd/673rd/674th/675th/676th/677th/678th/679th/680th/681st/682nd/683rd/684th/685th/686th/687th/688th/689th/690th/691st/692nd/693rd/694th/695th/696th/697th/698th/699th/700th/701st/702nd/703rd/704th/705th/706th/707th/708th/709th/710th/711st/712nd/713th/714th/715th/716th/717th/718th/719th/720th/721st/722nd/723rd/724th/725th/726th/727th/728th/729th/730th/731st/732nd/733rd/734th/735th/736th/737th/738th/739th/740th/741st/742nd/743rd/744th/745th/746th/747th/748th/749th/750th/751st/752nd/753rd/754th/755th/756th/757th/758th/759th/760th/761st/762nd/763rd/764th/765th/766th/767th/768th/769th/770th/771st/772nd/773rd/774th/775th/776th/777th/778th/779th/780th/781st/782nd/783rd/784th/785th/786th/787th/788th/789th/790th/791st/792nd/793rd/794th/795th/796th/797th/798th/799th/800th/801st/802nd/803rd/804th/805th/806th/807th/808th/809th/810th/811st/812nd/813th/814th/815th/816th/817th/818th/819th/820th/821st/822nd/823rd/824th/825th/826th/827th/828th/829th/830th/831st/832nd/833rd/834th/835th/836th/837th/838th/839th/840th/841st/842nd/843rd/844th/845th/846th/847th/848th/849th/850th/851st/85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量化交易是什么？

定义： 量化交易是把投资逻辑转化为可由数据检验、程序执行和风险规则约束的系统化决策过程。

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区别： 普通投资主要依赖人的主观判断，量化交易主要依赖明确、可检验、可复现的规则。

为什么要进行量化交易？

持续交易能学会盈利吗？

Day Trading for a Living?

Chague, De-Losso & Giovannetti (2020)

追踪 2013–2015 年进入巴西股指期货市场的个人交易者，重点研究坚持日内交易至少 300 天的 1,551 人。

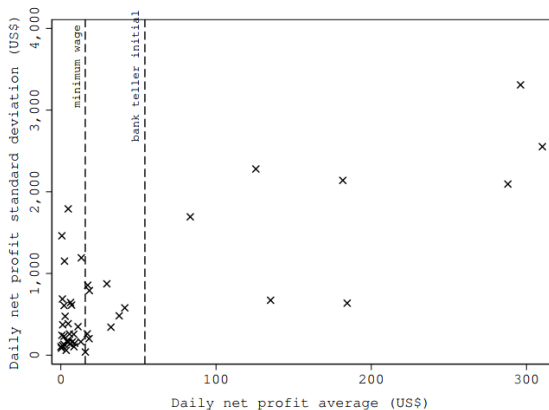
核心发现：97% 的长期交易者亏损；研究没有发现交易经验增加能够改善盈利表现的证据。

启示：

炒股实践长并无法保证盈利。

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3423101

少数盈利者?



模拟交易能有用吗？

Trading Simulations and Real Money Outcomes

Anginer, Piza, Ray & Xu (2020)

连接巴西投资者的模拟交易记录与真实账户表现，考察模拟经验是否带来更好的真实交易结果。

核心发现：模拟交易越活跃、承担风险越多的用户，更可能进入真实市场；但他们随后交易更频繁，真实收益也更差。

启示：

缺少正确评价和风险教育时，模拟交易可能强化过度自信，而不是提高能力。

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3634775

交易越活跃，真实表现越差？

Table 5: Univariate analysis of simulator account holders' real money account trading characteristics.

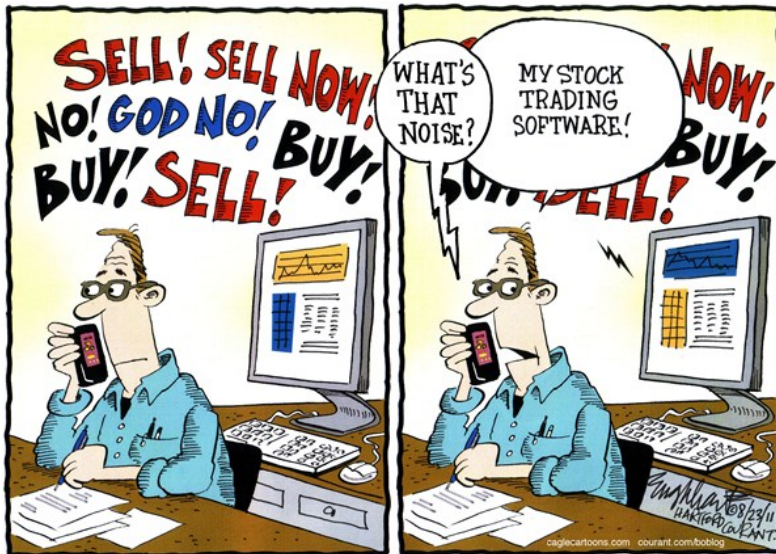
This table reports cross-sectional univariate analysis of simulator account holders' first three-month real money account trading characteristics after opening based on their last three-month simulator account trading characteristics. For each trading measure we divide all simulator accounts into two groups: High has values higher than the median while Low has the values lower than the median. We calculate the simulator account holders' actual trading account characteristics for the two simulator groups separately and test the difference between the two groups. For example investors with higher than the median number of trades in the simulator had 12.212 trades in their real money accounts.

(N=3,109)	High Group	Low Group	Difference	T-Statistics
Portfolio Market-Excess Return (% Monthly) (Real money account):				
No. of Total Trades (Simulator)	-1.183	-0.271	-0.912**	(2.327)
No. of Stocks Traded (Simulator)	-1.282	-0.155	-1.127***	(2.876)
Turnover (Monthly)(Simulator)	-0.293	0.182	-0.474	(0.827)
STD of Returns (% Monthly) (Simulator)	-0.728	-0.711	-0.017	(0.043)
Avg Trade Size (R\$) (Simulator)	-0.724	-1.381	0.657	(1.267)

模拟阶段交易越活跃、交易股票越多，真实账户的月度超额收益反而越低。

为什么要进行量化交易？

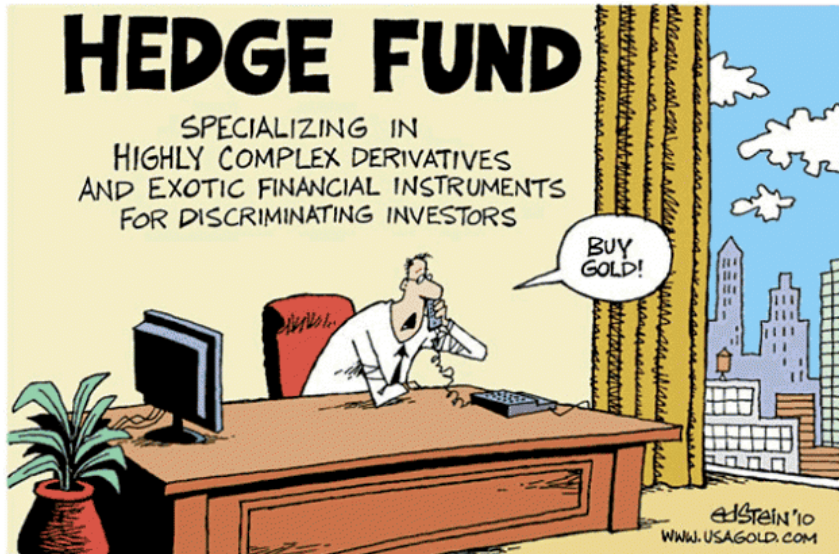
因为仅靠主观判断难以持续盈利，量化让投资逻辑变得可检验、可复现、可约束。



量化交易误区



"YOUR FATHER WAS RESPECTABLE, DEAR, BACK IN THE DAYS WHEN IT ACTUALLY MEANT SOMETHING TO HAVE A SEAT ON THE NEW YORK STOCK EXCHANGE."

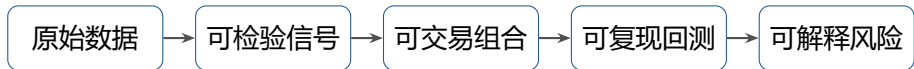


- ❶ **自动化可以提高执行一致性，但不等于可靠：**程序只会忠实执行规则，错误或冲突的信号同样会被快速执行。
- ❷ **量化交易依赖系统化研究，而不只是“用电脑炒股”：**技术降低了市场参与门槛，却没有降低交易研究、纪律和风险管理的难度。
- ❸ **复杂模型能够提升表达能力，但不必然带来有效策略：**复杂术语或许可以塑造专业形象，却无法替代深入的投资逻辑与理性判断。

目录

- ① 课程内容与基本信息
- ② 量化交易是什么
- ③ 量化交易的发展与生态
- ④ 投资标的与金融市场基础
- ⑤ 课堂实践

课程目标



核心目标

把一个投资想法转化为数据口径明确、策略规则清楚、回测结果可信、风险边界可解释的量化决策系统。

数据与收益

行情清洗、复权与时间对齐；收益率、回撤、风险和绩效指标。

因子与选股

风格因子构建、预处理、IC 与分层回测、横截面回归、多因子合成。

定价与组合

CAPM、CML 与 SML；均值-方差、有效前沿、组合约束与优化。

系统与风险

信号、持仓、交易与结算；交易成本、绩效评估、风险暴露与压力测试。

量化模型是交易信号的建模工具，但必须接受时间序列验证、真实回测和风险检验。

一个可复现的量化策略

从研究问题出发，完成数据处理、因子或模型构建、组合形成、真实回测、绩效评估和风险分析。

- 明确数据来源、时间范围与处理口径；
- 给出可执行的信号、调仓与持仓规则；
- 计入手续费、滑点和现实交易约束。
- 与合理基准比较收益和风险；
- 检验稳健性并讨论失效场景；
- 提交可复现代码、研究报告和展示。

评价原则

收益曲线是否漂亮固然重要，但项目质量取决于逻辑、证据和研究规范。

课程基本信息

项目	安排
学分学时	3 学分, 54 学时
主要工具	Python: anaconda 或 VS Code
主要数据	股票行情、财务、行业、指数与因子数据
学习方式	课堂教学、代码实践、阶段任务和综合项目
课程产出	可复现代码、策略研究报告与展示答辩

实践要求

每位同学都应能够独立运行课程代码、解释关键输出，并说明回测假设和风险。课程最后阶段让同学们答辩展示结果（先到先得）。

课程考核方式

- ① **平时成绩 (20%)**: 包括考勤、课堂提问与讨论表现、课堂小测等。
- ② **编程作业与阶段性项目 (20%)** 包括若干次上机实验报告, 以及 1-2 个阶段性小项目, 例如因子构建、简单策略回测等。
- ③ **期末综合项目 (60%)** 以 2-4 人小组为单位, 围绕近 3 年的一份量化研究报告或自拟题目, 完成策略复刻与扩展。

期末项目评价

课堂展示与答辩、同伴互评共同构成最终评分; 回测效果是重要评价维度之一, 需要与沪深 300 指数及基准多因子模型进行比较。

- ① **课程课件：**每次课的理论讲义与课堂 slides。
- ② **课程代码：**Python 示例程序、Spyder 源代码与上机实验模板。
- ③ **示例数据：**股票行情、指数、行业、财务与因子数据，以及字段说明。
- ④ **阅读材料：**量化研究报告、参考论文与阶段性项目说明。

券商	公众号	适合关注的内容
广发证券	广发金融工程研究 (gfquant)	因子、择时、衍生品与量化策略
华泰证券	华泰证券金融工程	因子研究、资产配置、机器学习、基金与 ETF
国信证券	量化藏经阁 (Z-quant)	因子选股、组合优化、基金研究
招商证券	招商定量任瞳团队 (RTQUANT)	多因子模型、行业轮动、基金评价
海通证券	海通量化团队 (ht_quant)	选股因子、高频数据、量化基金
方正证券	春晓量化 (foundersc_quant)	风格择时、多因子、ETF 与基金研究
中泰证券	量化先行者	因子选股、资产配置、基金研究
国盛证券	留富兵法 (fubing-quant)	资产配置、行业轮动、可转债
东吴证券	东吴金工与产品研究	量化选股、基金与 ETF、产品研究
华创证券	华创金工	择时、行业轮动、因子与事件研究

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量化交易的工作定义

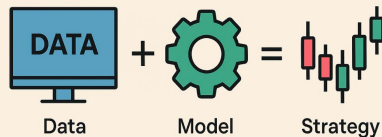
定义： 量化交易是把投资逻辑转化为可由数据检验、程序执行和风险规则约束的系统化决策过程。

关键词

- 数据驱动
- 规则明确
- 可回测
- 可执行
- 可复盘

量化交易是可验证、可重复、可监控的决策系统，而不是用电脑炒股。

What Is Quantitative Trading?



主观投资与量化投资的区别

关键区别在于投资判断如何形成、检验和执行，而不在于是否用电脑。

维度	主观投资	量化投资
信息来源	研究报告、调研、经验判断	结构化数据、因子、模型输出
决策方式	投资经理综合判断	规则或模型生成信号
覆盖范围	通常覆盖有限标的	可覆盖大量标的和长历史样本
可重复性	依赖个人经验与状态	规则固定时结果可复现
主要风险	情绪、认知偏差、风格漂移	数据错误、过拟合、模型失效

量化与传统投资可以互补，而非简单对立

人的优势

- 提出有经济意义的问题
- 理解制度、公司与市场环境
- 处理新事件和结构变化
- 判断模型边界与失败原因

系统的优势

- 一致处理大量数据和标的
- 快速检验历史证据
- 严格执行已定义规则
- 持续记录、监控和复盘

更准确的理解

量化投资是让我的投资逻辑接受数据检验并通过系统稳定执行，而不是让模型替代人。

量化策略如何分类？

策略逻辑

趋势跟踪
反转
统计套利
多因子
做市

市场暴露

方向性
相对价值
市场中性

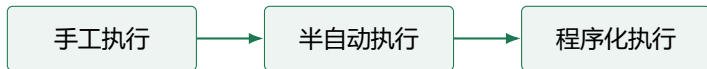
交易频率

低频
中频
高频

三个维度相互独立：一种策略需要同时考虑上述的三个维度。

从分类维度到具体策略

策略	策略逻辑	市场暴露	交易频率
期货 CTA	趋势跟踪	方向性	中频
市场中性选股	多因子	市场中性	低频或中频
高频做市	做市	低方向暴露	高频



区别主观交易和量化交易的依据是决策逻辑能否被检验与复现，而不是执行方式；量化交易也可以由交易员手工执行，程序化交易并不限于高频交易。

从投资问题到系统化决策

投资问题

- ① 买什么?
- ② 什么时候买?
- ③ 买多少?
- ④ 什么时候卖?
- ⑤ 如何控制风险?

量化表达

- ① 股票池与因子信号
- ② 调仓频率与交易规则
- ③ 权重约束与组合优化
- ④ 卖出规则与再平衡
- ⑤ 风险暴露、回撤与止损

量化交易的发展逻辑

- **数据可得性提高**：行情、财务、公告、指数、行业、因子库逐步标准化。
- **计算能力提升**：大样本回测、组合优化、机器学习模型变得可操作。
- **交易基础设施完善**：交易接口、风控系统、执行算法支持更稳定的策略落地。
- **投资过程可工程化**：研究、回测、交易、监控和复盘可以形成流水线。

重要判断

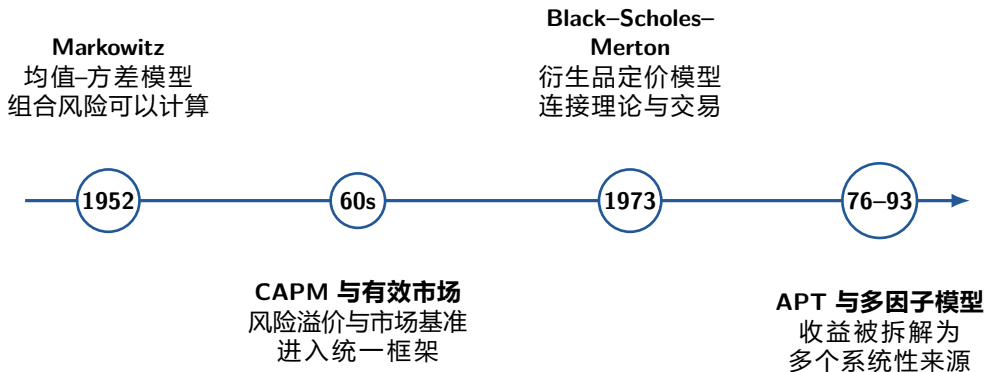
不确定性无法通过量化手段消灭，但可以使用纪律化流程管理不确定性。

这些条件并非同时出现：量化交易是在金融理论、计算技术和市场制度的共同演进中逐步形成的。

本讲位置

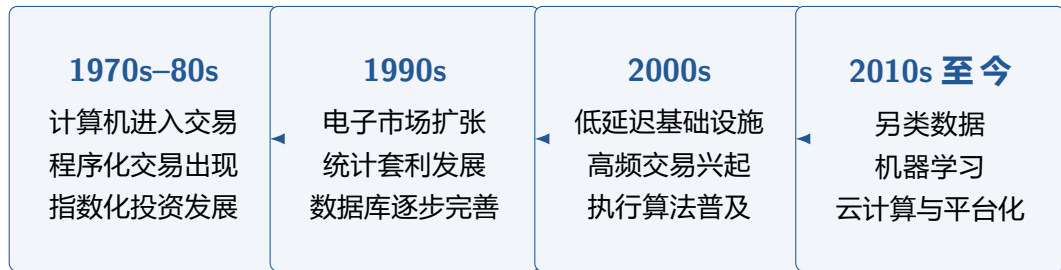
- ① 课程内容与基本信息
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量化投资的思想起点：把投资问题变成可计算问题



核心变化：投资从“判断一只资产”逐步变成“估计、比较并组合一组风险”。

从计算机交易到数据驱动投资



量化交易的演进：订单的执行速度越来越快，研究对象、数据范围和工程能力不断扩展。

中国量化投资的发展

上证 50 ETF 上市
境内指数化投资工具
开始形成

ETF 与衍生品扩展
指数增强、套利和
风险管理策略发展

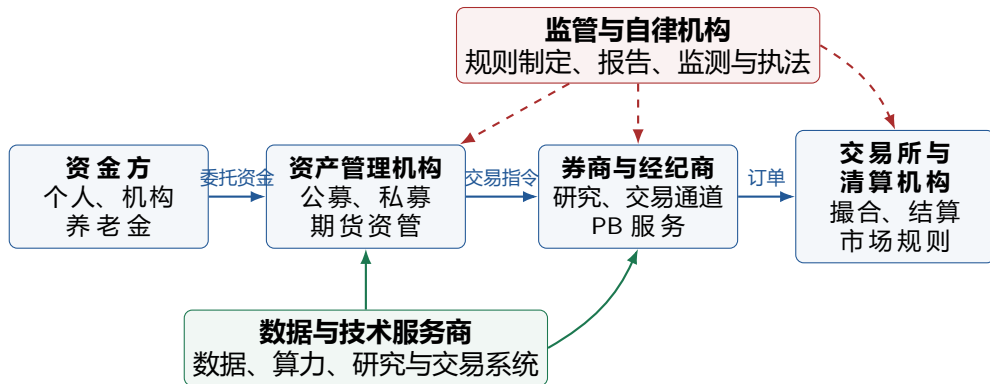


融资融券试点
沪深 300 股指期货
做空与对冲工具扩展

程序化交易监管完善
报告、监测、风控与
高频交易管理制度化

中国量化投资的发展始终受到可交易工具、市场制度与监管边界的共同影响。

资料来源：上海证券交易所、中国证监会、中国金融期货交易所。



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一个策略是在特定市场中交易具有特定权利、风险和生命周期的金融工具，而不是抽象地预测“价格”。

- ① **交易什么？** 股票、债券、商品、外汇等基础资产。
- ② **通过什么交易？** 现货、基金、ETF、期货、期权等金融工具。
- ③ **在哪里交易？** 一级市场、二级市场、交易所或场外市场。
- ④ **谁在参与？** 买方、卖方与市场基础设施。
- ⑤ **标的如何变化？** 发行、交易、公司行为、到期或退出。

我们究竟在交易什么？

常见基础资产：

类别	代表的经济关系	典型对象	核心问题
股票	对企业剩余收益和资产的所有权	个股、股票指数	企业价值是多少？
债券	发行人承诺支付的合同现金流	国债、金融债、信用债	现金流能否按期支付？
商品	对实物商品价格的风险暴露	能源、农产品、工业金属、贵金属	供需如何变化？
外汇	一种货币相对于另一种货币的价格	人民币、美元、欧元等货币对	相对购买力如何变化？

金融工具：是建立资产暴露的一类金融产品，包括衍生品、ETF 等。

从基础资产到金融工具

基础资产：风险和收益的来源

- 股票
- 债券
- 商品与贵金属
- 外汇



金融工具：控制风险和收益的手段

- 现货与证券
- 基金与 ETF
- 期货与远期
- 期权与互换

衍生品

建立在股票、债券、商品、外汇甚至其他衍生品之上的金融工具，而不是与这些基础资产互斥的第五类资产。

资产类别存在属性差异：模型须建立在标的性质的基础之上。

类别	主要收益来源	核心风险	典型量化问题
股票	盈利增长、估值变化、分红	市场、行业与公司风险	因子选股、指数增强
债券	票息、利率变化、信用利差	久期、信用与流动性风险	收益率曲线、利差交易
商品	现货变化、期限结构、展期收益	供需、库存、季节与地缘风险	趋势、期限结构策略
外汇	汇率变化、利差与套息收益	宏观政策、资本流动与尾部风险	动量、套息、宏观策略
衍生品	底层资产变化、波动率与时间价值	杠杆、非线性、到期与对手方风险	对冲、套利、波动率交易

本课程主要使用股票数据建立模型，但量化交易不仅限于股票交易，且股票量化交易模型不能不加修改地应用在其他资产类别上。

市场结构：一级市场与二级市场

一级市场：形成证券

- 新证券被发行和销售。
- 资金通常流向发行人。
- 例子：IPO¹、增发、债券发行。
- 核心功能：资本形成。

二级市场：转让证券

- 已发行证券在投资者之间交易。
- 发行人通常不获得新增资金。
- 例子：股票、债券、ETF² 交易。
- 核心功能：流动性与价格发现。

大多数量化交易发生在二级市场，但证券的发行条件会影响其估值、流动性与后续表现。

¹IPO: Initial Public Offering, 首次公开募股。

²ETF: Exchange-Traded Fund, 交易型开放式指数基金。

交易场所：交易所市场与 OTC 市场

维度	交易所市场	OTC 市场 ¹
组织方式	集中交易并遵循统一规则	通过双边协商或交易商网络交易
标准化	合约、报价单位和交易规则较统一	条款可以定制，结构更灵活
透明度	报价和成交信息通常更透明	交易前信息通常更分散
对手方与清算	通常通过中央对手方集中清算	更依赖交易对手信用与双边安排
典型工具	股票、ETF、场内期货与期权	债券、外汇、远期、互换与定制衍生品

交易所是一种市场组织与基础设施，不等同于某一类投资标的。

¹OTC: Over-the-Counter, 场外市场。

买方、卖方与市场基础设施

买方：管理资金并承担组合风险

- 公募基金、私募证券基金
- PE¹、VC²、保险、养老金与银行理财
- 目标是形成投资组合并实现风险收益目标

卖方：提供金融产品与市场服务

- 证券公司、投资银行、交易商
- 研究、承销、做市、经纪、交易与 PB³ 服务
- 收入主要来自服务、价差和业务费用

市场与信息基础设施

交易所、清算机构、登记托管机构、指数服务提供商⁴和监管部门共同提供交易组织、结算、指数编制与市场规则。

“买方/卖方”是机构功能的划分，而不是某一笔交易中的买入者和卖出者。

¹PE: Private Equity, 私募股权投资。

²VC: Venture Capital, 风险投资。

³PB: Prime Brokerage, 主经纪商服务, 面向对冲基金等专业机构, 整合交易、融资融券、托管清算与风险报告。

⁴指数服务提供商, 指数编制机构, 提供指数编制与发布服务。

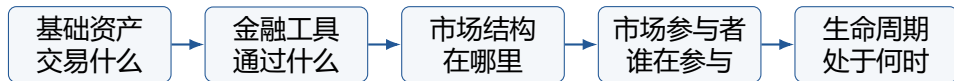
金融工具的生命周期

工具	形成	交易阶段	存续期事件	终点
股票	IPO、增发	上市后二级交易	分红、拆股、配股、并购	退市、私有化或并购退出
债券	债券发行	场内或 OTC 交易	付息、评级变化、回售赎回	到期兑付或违约
基金	募集、注册与成立	申购赎回或场内交易	分红、份额折算、组合调整	清盘、到期或转型
衍生品	交易所挂牌或场外签约	场内或 OTC 交易	保证金、盯市、行权、移仓	平仓、行权、交割或到期

量化研究中的含义

标的集合和价格序列都会随时间变化；股票复权、基金份额变化、债券剩余期限以及衍生品到期与移仓，都不能被当作普通缺失值处理。

从市场基础进入量化研究



回答这五个问题之后，才能进一步讨论如何获取数据、构造信号、形成组合并检验策略。

本讲位置

- ① 课程内容与基本信息
- ② 量化交易是什么
- ③ 量化交易的发展与生态
- ④ 投资标的与金融市场基础
- ⑤ 课堂实践

任务

使用 BaoStock API 下载沪深 300 (CSI 300) 指数日线数据，并实现一个 20 日均线择时策略。

- ① 安装 Anaconda，配置并启动 Spyder。
- ② 在 Spyder 中安装 BaoStock，建立 API 会话。
- ③ 下载并保存 sh.000300 的日线数据。
- ④ 计算 20 日移动平均线并生成择时信号。
- ⑤ 根据均线信号计算策略收益和净值。
- ⑥ 比较买入持有与均线策略的净值曲线。

配置 Anaconda 与 Spyder

下载安装程序

下载地址: <https://mirrors.tuna.tsinghua.edu.cn/anaconda/archive/>
指定版本: Anaconda3-2024.10-1-Windows-x86_64.exe

- ① 运行安装程序, 按默认选项完成安装。
- ② 从开始菜单启动 **Spyder**。
- ③ 选择 **Projects** → **New Project** → **Existing directory**。
- ④ 选择当前课程文件夹, 新建 Project, 并在其中建立 `code/lec01.py`。

环境检查

```
import numpy as np # 数值计算
import pandas as pd # 表格数据处理
np.__version__, pd.__version__ # 检查版本
```

在 Spyder 中安装并检查 BaoStock

在 Anaconda Prompt 中安装

```
pip install baostock -i https://pypi.org/simple # 安装数据接口
```

spyder 中新建 lec01.py 脚本并运行

```
import baostock as bs # 行情数据接口
import pandas as pd # 表格数据处理
import matplotlib.pyplot as plt # 绘图
```

```
login = bs.login() # 登录 BaoStock
assert login.error_code == "0", login.error_msg # 检查登录状态
print("BaoStock login success") # 输出登录结果
```

下载沪深 300 日线数据 (一): 设置参数

```
fields = "date,code,open,high,low,close,preclose,volume,amount" # 返回字段  
  
start_date = "2015-01-01" # 样本开始日期  
end_date = pd.Timestamp.today().strftime("%Y-%m-%d") # 当前日期
```

- fields: 接口返回的日期、代码、价格、成交量与成交额字段。
- start_date/end_date: 样本的开始日期和结束日期。
- frequency="d": 下载日线数据。
- adjustflag="3": 不复权; 指数本身不存在股票分红复权问题。

下载沪深 300 日线数据 (二): 调用接口

```
rs = bs.query_history_k_data_plus(  
    "sh.000300", # 沪深300代码  
    fields, # 指定返回字段  
    start_date=start_date, # 开始日期  
    end_date=end_date, # 结束日期  
    frequency="d", # 日线频率  
    adjustflag="3", # 不复权  
)  
assert rs.error_code == "0", rs.error_msg # 检查接口状态  
  
rows = [] # 存放逐行返回的数据  
while rs.next(): # 遍历查询结果  
    rows.append(rs.get_row_data()) # 保存当前记录  
  
df = pd.DataFrame(rows, columns=rs.fields) # 转为 DataFrame  
bs.logout() # 退出接口
```

代码说明: sh.000300 是沪深 300 的上海行情代码; 接口返回的数据先逐行加入 rows, 再转换为 DataFrame。

整理、检查并保存数据

```
from pathlib import Path # 处理文件路径

df["date"] = pd.to_datetime(df["date"]) # 转换日期类型
price_cols = ["open", "high", "low", "close", "preclose"] # 价格列
df[price_cols] = df[price_cols].apply(
    pd.to_numeric, errors="coerce"
) # 转为数值, 异常值记为缺失

df = (df.sort_values("date")
      .drop_duplicates("date")
      .dropna(subset=["close"])) # 排序、去重并删除缺失价格

print(df.head()) # 查看前几行
print(df.isna().sum()) # 统计缺失值

Path("data/raw").mkdir(parents=True, exist_ok=True) # 创建目录
df.to_csv("data/raw/csi300_daily.csv", index=False) # 保存数据
```

原始下载结果先保存, 再进行策略计算, 便于复现并减少重复调用 API。

绘制沪深 300 与 20 日均线

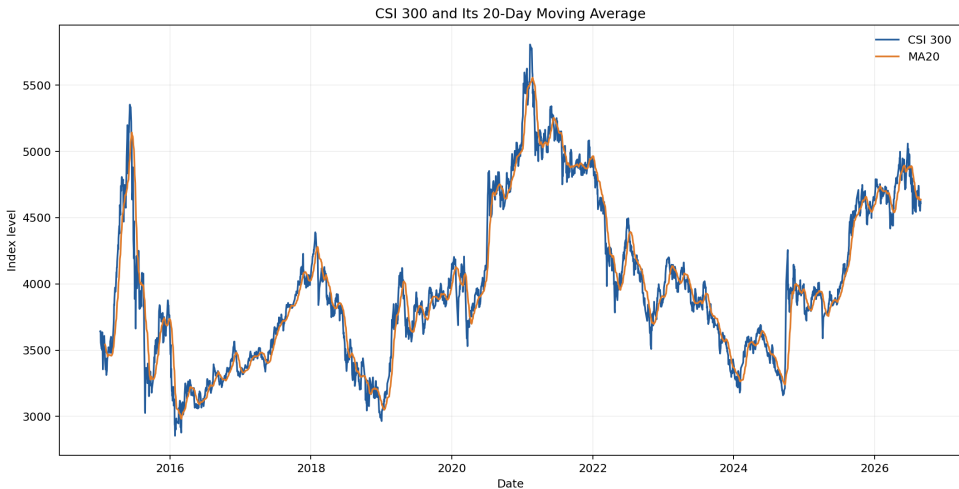
```
df["ma20"] = df["close"].rolling(20).mean() # 计算20日均线

fig, ax = plt.subplots(figsize=(12, 6.2)) # 创建画布
ax.plot(df["date"], df["close"], label="CSI 300") # 指数曲线
ax.plot(df["date"], df["ma20"], label="MA20") # 均线曲线
ax.set_title("CSI 300 and Its 20-Day Moving Average")
ax.set_xlabel("Date")
ax.set_ylabel("Index level")
ax.grid(alpha=0.2) # 添加网格
ax.legend() # 显示图例

plt.tight_layout() # 自动调整边距
plt.show() # 显示图形
```

观察任务：价格与均线的交叉分别出现在什么市场阶段？

绘图结果：沪深 300 与 20 日均线



策略规则：20 日均线择时

$$MA_{20,t} = \frac{1}{20} \sum_{j=0}^{19} P_{t-j}, \quad S_t = \mathbb{I}(P_t > MA_{20,t}).$$

$$R_t^{\text{strategy}} = S_t R_t^{\text{CSI300}}.$$

- 收盘价高于 20 日均线：当日持有沪深 300。
- 收盘价不高于 20 日均线：当日持有现金，现金收益暂设为零。
- 策略只做多，不使用杠杆，暂不计交易成本。

按照策略规则计算净值

```
df["ma20"] = df["close"].rolling(20).mean() # 计算20日均线
df["signal"] = (df["close"] > df["ma20"]).astype(int) # 生成信号

df["position"] = df["signal"] # 当日信号作为当日仓位

df["index_ret"] = df["close"].pct_change() # 计算指数收益率
df["strategy_ret"] = df["position"] * df["index_ret"] # 策略收益

df["index_nav"] = (1 + df["index_ret"].fillna(0)).cumprod() # 指数净值
df["strategy_nav"] = (
    1 + df["strategy_ret"].fillna(0)
).cumprod() # 策略净值
```

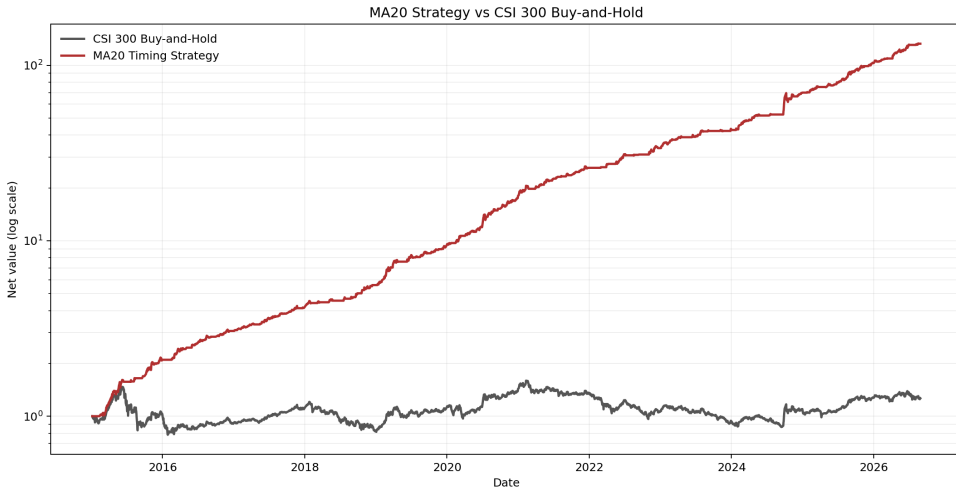
用对数坐标比较策略与 Buy-and-Hold

```
fig, ax = plt.subplots(figsize=(12, 6.2)) # 创建画布
ax.plot(df["date"], df["index_nav"],
        label="CSI 300 Buy-and-Hold") # 买入持有净值
ax.plot(df["date"], df["strategy_nav"],
        label="MA20 Timing Strategy") # 择时策略净值
ax.set_yscale("log") # 使用对数纵轴
ax.set_ylabel("Net value (log scale)")
ax.set_xlabel("Date")
ax.grid(alpha=0.2, which="both") # 添加网格
ax.legend() # 显示图例

plt.tight_layout() # 自动调整边距
plt.show() # 显示图形
```

当前比较忽略手续费、滑点和现金利息，因此只是教学用的最小回测。

回测结果：MA20 策略与 Buy-and-Hold



效果如此之好!

这个结果可信吗？

为什么一个如此简单的规则，表现却可能好得出奇？

讨论：

请先不要修改代码。检查策略规则、信号生成和收益计算的时间顺序，找出可能的问题。

- t 日收益在什么时候才能完全确定？
- t 日收盘价在什么时候才能观察到？
- 使用收盘价生成的信号，最早可以在什么时候执行？

陷阱：信号使用了未来信息

错误在哪里？

S_t 需要 t 日收盘价才能计算，却被用于获取从 $t-1$ 日收盘到 t 日收盘的收益。策略在收益发生之后，反过来决定是否持有。

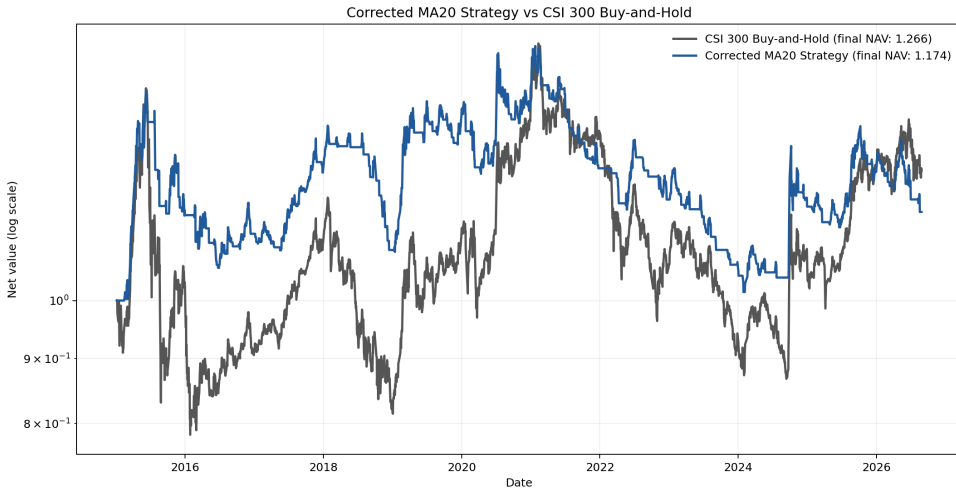
正确的时间关系是

$$R_{t+1}^{\text{strategy}} = S_t R_{t+1}^{\text{CSI300}}.$$

```
df["position"] = df["signal"].shift(1).fillna(0) # 信号滞后一日
df["strategy_ret"] = df["position"] * df["index_ret"] # 可实现收益
df["strategy_nav"] = (
    1 + df["strategy_ret"].fillna(0)
).cumprod() # 修正后的策略净值
```

重新画图：比较修正前后的净值曲线，观察“一个交易日”的错位能造成多大的虚假优势。

修正后的回测结果



真实可执行的策略：使用滞后一日的信号后，MA20 策略与 CSI 300 Buy-and-Hold 的差异显著缩小。

结果检查与讨论

- ❶ 修正未来函数之后，策略净值发生了多大变化？
- ❷ 20 日均线策略在哪些市场阶段持有现金？它规避了哪些下跌，又错过了哪些上涨？
- ❸ 把窗口改为 10、60 或 120 日，结果如何变化？
- ❹ 如果每次仓位变化收取交易成本，策略结果会如何改变？

提交结果

API 下载代码、保存的 CSV、价格与均线图、两条净值曲线，以及对结果的三句话解释。

课后作业：构建上证股票历史数据集

任务

使用 BaoStock 下载 2015-01-01 至 2025-12-31 的上证 A 股日线数据，为下一讲的课堂练习准备数据集。

- ① 用 `query_stock_basic()` 获取证券列表，保留 `type="1"` 且代码以 `sh.6` 开头的股票。
- ② 保存 `ipoDate`、`outDate` 和 `status`，需要保留上市及退市的股票。
- ③ 下载字段：日期、代码、OHLC、前收盘价、成交量、成交额、换手率、交易状态、涨跌幅和 ST 标记。
- ④ 每只股票单独保存为 `data/raw/sse_daily/<code>.csv`。

本讲小结

- 量化交易把投资逻辑转化为由数据检验、程序执行和风险规则约束的系统化决策过程。
- 主观投资与量化投资使用的工具和决策方式不同，但都需要经济逻辑、信息判断与风险意识。
- 量化交易的发展来自数据、计算能力、市场电子化和机构化需求，其生态包括买方、卖方与市场基础设施。
- 理解股票、债券、商品、外汇、基金和衍生品，以及一二级市场与证券生命周期，是设计策略的市场基础。
- 课堂实践使用 Spyder 和 BaoStock 下载沪深 300 日线数据，绘制指数及其 20 日均线，并完成一个最小择时回测。
- 回测必须遵守信息可得时间：收盘后生成的信号只能用于下一交易日，否则会因未来函数产生虚假收益。
- 课后将构建包含退市股票的上证历史行情数据集，为下一讲研究小市值策略的可实现性做准备。